ETAS RTA-BSW v12.7.0

RTA-Hwd (Lin) Reference Guide

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1 Lin Transceiver

Module name	LinTrcv
Package	RTA-HWD (Lin)
AUTOSAR Module ID	64
AUTOSAR Version	23-11

1.1 Introduction

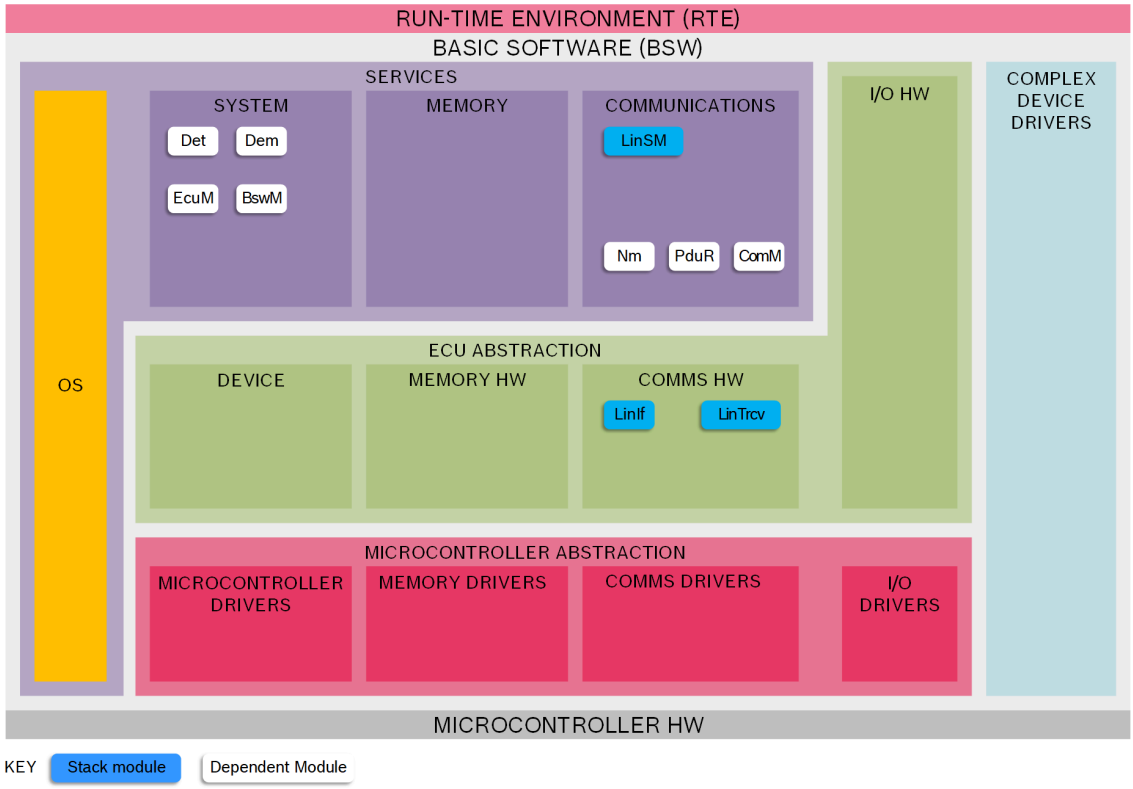

Figure 1.1.RTA-LIN Architecture

The LIN Transceiver Driver (LinTrcv) is located in the communication hardware extraction layer, between the communication drivers layer and the upper communication services layer, and it forms part of the RTA-HWD (LIN) stack.

The diagram above shows the position of the LinTrcv module in the AUTOSAR layered architecture.

LinTrcv abstracts the different LIN hardware transceivers to a standardised interface for the upper layer LIN Interface (LinIf). It uses MCAL to access the hardware, and adjustment for a specific hardware transceiver is done within the configuration of LinTrcv.

The functionality provided by LinTrcv differs for each hardware transceiver - see the Supported Features section for more details.

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

1.2 Supported Features

1.2.1 General

- Getting/Setting the mode of the transceiver
- Getting/Setting the wakeup mode of the transceiver
- Transmitting and receiving data via the LIN bus

1.2.2 Wakeup Detection

LinTrcv only partially supports Wakeup Detection. See the Limitations section for details.

1.2.3 Supported Hardware Devices

LinTrcv supports the following hardware devices:

Manufacturer	Device Name
NXP Semiconductors	TJA1021
NXP Semiconductors	TJA1027
NXP Semiconductors	SIP2x0_TJA1901
NXP Semiconductors	TJA1902
NXP Semiconductors	TJA1029
ON Semiconductors	NCV7424
ON Semiconductors	NCV7321
Infineon	TLE7259
Texas Instruments	TLIN1021A
Texas Instruments	TLIN1024A

1.2.3.1 NCV7424

The following points should be noted with regard to the NCV7424 transceiver.

- The NCV7424 has 4 LIN channels but there are no separate mode control Pins to control the modes of LinTrcv channels separately.
- During each driving cycle, the common state should be maintained across the LinTrcv channels.
- If wakeup is detected on any one channel, then all the channels will become active irrespective of the current state of other LinTrcv channels.

1.2.3.2 TLIN1024A

The following point should be noted with regard to the TLIN1024A transceiver.

- The TLIN1024A transceiver supports 4 Lin channels with separate enable pins.

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 LinTrcv/LinTrcvChannel/LinTrcvAccess/LinTrcvDioAccess/LinTrcv-DioChannelAccess/LinTrcvDioHardwareInterfaceType

Short Name	LinTrcvDioHardwareInterfaceType
Description	Type of functionalty of this pin. The trcv-software uses this parameter to know what functionality is located at the referenced Dio Pin conneted to the Trcv-HW.
Multiplicity	1..1
Default Value	Enable
Enumeration Literals	Enable, WakeupSignalisation
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.2 LinTrcv/LinTrcvChannel/LinTrcvAccess/LinTrcvDioAccess/LinTrcv-DioChannelAccess/LinTrcvRbDioSymName

Short Name	LinTrcvRbDioSymName
Description	This parameter have to be configured only if Dio component is not available in the Project. In this case Macro to access Dio will be generated based on the name given here. Macro is generated by attaching the symbolic name pattern. If third party macro is not matching with the generated macro , then a mapping macro should by created by integrator. If project has Autosar Dio then no need to configure this parameter.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.3 LinTrcv/LinTrcvChannel/LinTrcvAccess/LinTrcvDioAccess/LinTrcv-

DioChannelAccess/LinTrcvRbHardwarePinInvert

Short Name	LinTrcvRbHardwarePinInvert
Description	Parameter depends on hardware: Describes, if a inverter is between transceiver pin and micro controller. If the transceiver is directly connected to the Controller, 'false' should be configured. If there is an inverter between transceiver and micro controller, 'yes' should be configured.
Introduction	false: No inverter between Micro Controller and transceiver yes: Inverter between Micro Controller and transceiver
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.4 LinTrcv/LinTrcvChannel/LinTrcvHardwareType

Short Name	LinTrcvHardwareType
Description	Type of the used Hardware
Introduction	ImplementationType: LinTrcv_TrcvModeType
Multiplicity	1..1
Default Value	-
Enumeration Literals	NCV7321, TJA1021, TJA1027, SIP2x0_TJA1901_TJA1902_TJA1029, TLE7259, NCV7424, TLIN1021A, TLIN1024A
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.5 LinTrcv/LinTrcvDemEventParameterRefs

Short Name	LinTrcvDemEventParameterRefs
Description	Container for the references to DemEventParameter elements which shall be invoked using the API Dem_SetEventStatus API in case the corresponding error occurs. The EventId is taken from the referenced DemEventParameters DemEventId value. The standardized errors are provided in the container and can be extended by vendor specific error references.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.6 LinTrcv/LinTrcvDemEventParameterRefs/LINTRCV_E_NO_TRCV_-CONTROL

Short Name	LINTRCV_E_NO_TRCV_CONTROL
Description	Reference to the DemEventParameter which shall be issued when the error "No/incorrect communication to transceiver" has occured.
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.7 LinTrcv/LinTrcvDemEventParameterRefs/LINTRCV_E_UNEXPECT-ED_MODE_PROP

Short Name	LINTRCV_E_UNEXPECTED_MODE_PROP
Description	Reference to the DemEventParameter which shall be issued when the unexpected mode propoga-tion is detcted in Transceiver HW. If this Parameter is configured then MainFunction period should be configured in parameter LinTrcvRbMainFunction-Period to monitor the unexpected mode transac-tion
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.8 LinTrcv/LinTrcvDemEventParameterRefs/LinTrcvRbDynCre-ate_LINTRCV_E_NO_TRCV_CONTROL

Short Name	LinTrcvRbDynCreate_LINTRCV_E_NO_TRCV_ CONTROL
Long Name	Dynamically create DemEvent
Description	Selects the way how the example DemEvent is created. If not set or set to "Disabled", no such DemEvent is automatically created. If the dynamic creation of this DemEvent is enabled, the reference to this DemEvent (in parameter rba_Example_DemEventRef) is also automatically created and shall therefore not be set manually.
Multiplicity	0..1
Default Value	Disabled
Enumeration Literals	Disabled, Standard
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.9 LinTrcv/LinTrcvGeneral/LinTrcvAutosarVersion

Short Name	LinTrcvAutosarVersion
Description	Naming convention for symbolic names are generated according autosar version: Lecacy: no prefix AR4.0.2: Dio_ (ecuc_sws_2108) AR4.0.3: DioConf_DioChannel_
Introduction	Naming convention for symbolic names are generated according autosar version: Lecacy: no prefix AR4.0.2: Dio_ AR4.0.3: DioConf_DioChannel_
Multiplicity	1..1
Default Value	-
Enumeration Literals	Legacy, AR4_0_2, AR4_0_3
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.10 LinTrcv/LinTrcvGeneral/LinTrcvClearWupReasonInSetWakeup-Mode

Short Name	LinTrcvClearWupReasonInSetWakeupMode
Description	On true, the wakeup reason is cleared in function LinTrcv_SetWakeupMode with Parameter LINTRCV_TRCV_WU_CLEAR. With other parameters or 'false' at this configuration parameter, the wakeup reason is not cleared in function LinTrcv_SetWakeupMode. The wakeup reason is cleard always init-function and on transition to sleep mode. If this parameter is set to true in one Lin Trcv it will be used in all LinTrcv
Introduction	True: WakeupReason is cleared False: WakeupReason is not cleared
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.11 LinTrcv/LinTrcvGeneral/LinTrcvInformEcumInCheckWakeup

Short Name	LinTrcvInformEcumInCheckWakeup
Description	If enabled, the EcuM EcuM_SetWakeupEvent is called in function LinTrcv_CheckWakeup if wakeup occured. The #define LINTRCV_ECUM_WAKEUP-CONFIGURAION_USED is generated according configuration value
Introduction	True: Is used False: Is not used
Multiplicity	1..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.12 LinTrcv/LinTrcvGeneral/LinTrcvInformLinIfInCheckWakeup

Short Name	LinTrcvInformLinIfInCheckWakeup
Description	This value should always be off! If enabled, the LinIf LinIf_WakeupConfirmationt is called in function LinTrcv_CheckWakeup if wakeup occured. The #define LINTRCV_LINIF_WAKEUPCONFIGURAION_USED is generated according configuration value.
Introduction	True: Is used False: Is not used
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.13 LinTrcv/LinTrcvGeneral/LinTrcvRbBswmdBooleanSize

Short Name	LinTrcvRbBswmdBooleanSize
Description	At BSWMD generation the bit-size of boolean type has to be forwarded to rba_Arxml_Gen and can be configured here.
Multiplicity	0..1
Default Value	LINTRCV_BSWMDVALUESIZE_8BIT
Enumeration Literals	LINTRCV_BSWMDVALUESIZE_1BIT, LINTRCV_BSWMDVALUESIZE_8BIT, LINTRCV_BSWMDVALUESIZE_16BIT, LINTRCV_BSWMDVALUESIZE_32BIT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.14 LinTrcv/LinTrcvGeneral/LinTrcvRbBswmdEnumSize

Short Name	LinTrcvRbBswmdEnumSize
Description	At BSWMD generation the bit-size of enums has to be forwarded to rba_Arxml_Gen and can be configured here.
Multiplicity	0..1
Default Value	LINTRCV_BSWMDVALUESIZE_8BIT
Enumeration Literals	LINTRCV_BSWMDVALUESIZE_8BIT, LINTRCV_BSWMDVALUESIZE_16BIT, LINTRCV_BSWMDVALUESIZE_32BIT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.15 LinTrcv/LinTrcvGeneral/LinTrcvRbBswmdSpiSize

Short Name	LinTrcvRbBswmdSpiSize

Description	At BSWMD generation the bit-size of Spi_DataType has to be forwarded to rba_Arxml_Gen and can be configured here.
Multiplicity	0..1
Default Value	LINTRCV_BSWMDVALUESIZE_16BIT
Enumeration Literals	LINTRCV_BSWMDVALUESIZE_8BIT, LINTRCV_BSWMDVALUESIZE_16BIT, LINTRCV_BSWMDVALUESIZE_32BIT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.16 LinTrcv/LinTrcvGeneral/LinTrcvRbMainFunctionPeriod

Short Name	LinTrcvRbMainFunctionPeriod
Description	This parameter describes the period for cyclic call to LinTrcv_RbMainFunction. Unit is seconds. This parameter need to be configured if DEM parameter LINTRCV_E_UNEXPECTED_MODE_PROP is configured
Multiplicity	0..1
Value Range	0.001..65535
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.17 LinTrcv/LinTrcvGeneral/LinTrcvUseAsr3Api

Short Name	LinTrcvUseAsr3Api
Description	Switches development error detection and notification on and off. If switched on, #define LINTRCV_DEV_ERROR_DETECT ON shall be generated. If switched off, #define LINTRCV_DEV_ERROR_DETECT OFF shall be generated. Define shall be part of file LinTrcv_Cfg.h.
Introduction	True: Is used False: Is not used
Multiplicity	1..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.2.1 LinTrcv_RbMainFunction

Syntax	`LinTrcv_RbMainFunction(void)`
Description	Cyclic function of transceiver driver.

Note	This function needs to be called from OS only if reporting of Dem in case of unexpected mode propogation is required.

1.3.2.2 LinTrcv_CheckModePropagation

Syntax	`LinTrcv_CheckModePropagation(uint8 LinNetwork)`
Parameters In	`LinNetwork` Identifies the Transceiver to which the API has to be applied.
Description	Checks Latest mode with the current mode of Transceiver.

1.3.2.3 LinTrcv_SetTransceiverMode

Syntax	`LinTrcv_SetTransceiverMode(uint8 LinTrcvId_u8, LinTrcv_TrcvModeType LinTrcv_TrcvMode)`
Parameters In	`LinTrcvId_u8` Identifies the Transceiver to which the API has to be applied. `LinTrcv_TrcvMode` Selects the state the transceiver will transit to (value points to table lintrcv_SetModeStructChain).
Description	Sets the transceiver mode.

1.3.2.4 LinTrcv_GetTransceiverMode

Syntax	`LinTrcv_GetTransceiverMode( uint8 LinTrcvId_u8, LinTrcv_TrcvModeType* LinTrcvModePtr )`
Parameters In	`LinTrcvId_u8` Identifies the Transceiver to which the API has to be applied.
Parameters Out	`LinTrcvModePtr` Current state of the transceiver.
Description	Returns the current state of the transceiver.

1.4 Limitations

1.4.1 Wakeup Detection

Wakeup Detection is only partially supported by LinTrcv. This is because the wakeup reason cannot currently be distinguished (only one wakeup reason is available). External wakeup is detected only when the time of calling LinTrcv_CheckWakeup and low phase on the LIN bus is matched.

1.4.2 Wakeup By Pin

Wake-Up by Pin is not implemented for LinTrcv.

1.4.3 LinTransceiver Initialization

A LinTransceiver that is not currently connected to hardware will be incorrectly set to initialized if the SLP_N pin is low. So, when configuring a dummy transceiver, ensure that the SLP_N pin is high during initiation.

1.4.4 Fast Programming (TLE7259)

Fast Programming mode on the TLE7259 is not supported.

1.5 Exclusions

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- LinTrcv/LinTrcvChannel/LinTrcvAccess/LinTrcvDioAccess/LinTrcvDioChannelAccess/LinTrcvHardwareInterfaceName
- LinTrcv/LinTrcvChannel/LinTrcvAccess/LinTrcvSpiSequence
- LinTrcv/LinTrcvChannel/LinTrcvAccess/LinTrcvSpiSequence/LinTrcvSpiSequenceName
- LinTrcv/LinTrcvChannel/LinTrcvChannelEcucPartitionRef
- LinTrcv/LinTrcvChannel/LinTrcvChannelUsed
- LinTrcv/LinTrcvGeneral/LinTrcvEcucPartitionRef
- LinTrcv/LinTrcvGeneral/LinTrcvTimerType
- LinTrcv/LinTrcvGeneral/LinTrcvWaitTime

1.6 Deviations

1.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

1.6.1.1 LinTrcv/LinTrcvChannel/LinTrcvChannelId

[RTA-BSW] Description	Unique identifier of the LIN Transceiver Channel. ! Parameter set with “Generate ID” !
[AUTOSAR] Description	Unique identifier of the LIN Transceiver Channel.

1.6.1.2 LinTrcv/LinTrcvChannel/LinTrcvIcuChannelRef

[RTA-BSW] Description	Reference to the IcuChannel to enable/disable the interrupts for wakeups.This Parameter is not used !
[AUTOSAR] Description	Reference to the IcuChannel to enable/disable the interrupts for wakeups.

1.6.1.3 LinTrcv/LinTrcvChannel/LinTrcvWakeupSourceRef

[RTA-BSW] Description	Reference to a wakeup source in the EcuM configuration. This reference is only needed if LinTrcvWakeupByBusUsed is true. Implementation Type: reference to EcuM_WakeupSourceType. ! Parameter not used !

[AUTOSAR] Description	Reference to a wakeup source in the EcuM configuration. This reference is only needed if LinTrcvWakeupByBusUsed is true. Implementation Type: reference to EcuM_WakeupSourceType.

1.6.1.4 LinTrcv/LinTrcvGeneral/LinTrcvIndex

[RTA-BSW] Description	Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0. ! Parameter not used !
[AUTOSAR] Description	Specifies the InstanceId of this module instance. If only one instance is present it shall have the Id 0.

1.7 API Definition

Table 1.28.LinTrcv Supported API functions

`LinTrcv_Init`	Initialises the module.
`LinTrcv_SetOpMode`	The internal state of the LIN transceiver driver is switched to the mode given in the OpMode parameter.
`LinTrcv_GetOpMode`	Detects the actual software state of LIN transceiver driver.
`LinTrcv_GetBusWuReason`	Provides the reason for the wakeup that the LIN transceiver has detected.
`LinTrcv_GetVersionInfo`	This service provides the version information of this module.
`LinTrcv_CheckWakeup`	Notifies the calling function if a wakeup is detected.
`LinTrcv_SetWakeupMode`	Enables/disables and clears the notification for wakeup events on the addressed network.

For further information on these functions, please refer to AUTOSAR_SWS_LIN-TransceiverDriver.pdf

1.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the “Configuration Generation” section of the RTA-CAR User Guide.

1.8.1 Transceiver Configuration

Transceivers are identified by a unique ID, which is set on a per-channel basis using the configuration parameters contained in `LinTrcvChannel`. Therefore, you should configure multi-channel transceivers with a separate set of parameters

for each channel (see Multi-Channel Transceivers below).

For transceivers which use the DIO interface, configure the DIO port for the port pin which is connected to the transceiver's Enable, Standby or Error pin. This information will be available in the vendor's ECU schematic. These settings are made under the `LinTrcvDioAccess` parameters.

1.8.1.1 Multi-Channel Transceivers

Some transceivers have multiple channels, and the name of these transceivers may not be found directly in the transceiver HW list. You should refer to the transceiver's user manual to identify the names of individual transceiver channels, which will need to be provided during configuration.

For example, the TJA1901 is actually: TJA1046 + TJA1029 (2 x TJA1044GT + TJA1029).

TJA1901 will not be available as a name in the transceiver list, but TJA1044 and TJA1029 will be available, and these should be selected instead of TJA1901.

1.8.1.2 DIO Port Configuration

Configure the DIO Port for the Port Pins that are connected to the transceiver's NSLP Pin. Check the ECU schematic for the Port Pin that is connected to the microcontroller.

1.9 Integration Advice

1.9.1 Init Functions

The Lin_Trcv module is initialized via LinTrcv_Init.

1.9.2 DeInit Functions

No specific handling necessary.

1.9.3 Scheduled Functions

LinTrcv_RbMainFunction is the cyclic function of LinTrcv.

1.9.4 LINTRCV_GETTIME

The integration file LinTrcv_Cfg_Time.h should be modified to provide `LINTRCV_GETTIME` and an appropriate definition of `LinTrcv_Cfg_Tick_to`.

`LINTRCV_GETTIME` should provide a time value from a free-running counter. The period of the counter must be less than or equal to 1 microsecond. `LINTRCV_CFG_TIME_FACTOR` should be set to the number of counter ticks per microsecond.

1.9.5 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in

the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The LinTrcv module uses the following OS resources:

- LinTrcv_RESOURCE

1.10 Hardware Requirements

Table 1.29.LinTrcv Hardware Requirements

Support	Usage
FLASH	428 bytes
RAM	9 bytes

Usage calculated based on a sample configuration.

2 Glossary

DIO

Digital Input/Output

LinIf

AUTOSAR LIN Interface module

LinSM

AUTOSAR LIN State Manager module

LinTp

AUTOSAR LIN Transport Layer module

MCAL

Microcontroller Abstraction Layer